Train a Human Neural Network

Name _____

Work with a partner. Compare answers when you finish each problem.

The club: Fighters and Wizards can join. Giants and Minis can't. **Strong** = 4 or 5 stars = **1**. Not strong = **0**.

Fire: a neuron stands up if its total is **more than** its threshold.

Fix rule: Should have fired: **ADD** 1 to each weight whose input was 1, threshold −1. Fired but shouldn't: **SUBTRACT**.

Who's to Blame? Find the hidden neuron closest to its threshold. Would the answer be right if it changed? Yes: it fixes itself. No: the output fixes itself.

1. Can one neuron do it? Try weights for one neuron. Test all 4 creatures.

Strong Strength weight _____ Strong Magic weight _____ Threshold _____ Right: _____ of 4

Did anyone get 4 of 4? _____ Why not? _____

2. Pass 1. Start with the numbers on the board. Write the totals for each card.

Card	Inputs	Hidden 1 total	Hidden 2 total	Output total	Says	Right?	Who fixes?
T1 Konik	1, 0						
T2 Pasir							
T3 Vasi							
T4 Skape							
T5 Porris							
T6 Galla							
T7 Gilax							
T8 Grapis							

Mistakes in pass 1: _____

3. The numbers on the board. After each fix, write all 9 numbers.

		Hidden 1			Hidden 2			Output		
	Card	Strength	Magic	thresh.	Strength	Magic	thresh.	Hidden 1	Hidden 2	thresh.
Start		−1	+1	1	0	−1	1	+1	+1	1
Fix 1										
Fix 2										
Fix 3										
Fix 4										
Fix 5										
Fix 6										

Word bank: Fighter Giant Mini Wizard

You can use a word more than once.

4. Pass 2. Same cards, same order. Use your numbers from the end of pass 1.

Card	Hidden 1 total	Hidden 2 total	Output total	Says	Right?
T1 Konik					
T2 Pasir					
T3 Vasi					
T4 Skape					

Card	Right?
T5 Porris	
T6 Galla	
T7 Gilax	
T8 Grapis	

Mistakes in pass 2: _____ (If you made mistakes, do another pass on the back.)

5. What did the hidden neurons learn?

Hidden 1 fires only for a _____. Hidden 2 fires only for a _____.

Did anyone tell them to look for these? _____ How did they learn it? _____

6. Test cards. Do **not** fix anything on a test!

Card	Stars	Inputs	Network says	Really (unfold)
X1 Kromos	5, 2			
X2 Vepul	2, 4			
X3 Tarra	5, 4			
X4 Volar	1, 2			
X5 Tibi	3, 2			
X6 Gobur	3, 5			

Why did the network get Tibi wrong? _____

7. Talk about it. One neuron couldn't learn the club rule because _____

_____

Go Further (Day 2). Same cards, different start. Use the tool on the lesson page.

Start B: did it learn? _____ Hidden 2 fires only for a _____. Start C: did it learn? _____

What happened with Start C? _____